

■ ESP32 Smart Grid Monitoring & Fault Analysis System

Complete University Master Textbook & Reverse Engineering Manual — Chapters 1 to 15

Chapter 3: Streamlit Architecture, Widget Lifecycle & Session State

3.1 Streamlit Core Architecture

Streamlit is an open-source Python framework that converts data scripts into web applications. It runs a Tornado web server and React frontend on port 8501, communicating via WebSockets.

3.2 Execution Lifecycle & Script Rerun Mechanism

Every time a user interacts with any widget (slider, button, radio menu), Streamlit re-executes the Python script from top to bottom. State is preserved using `st.session_state` or `@st.cache_resource`.

3.3 Memory Caching (@st.cache_resource vs @st.cache_data)

`@st.cache_resource` caches global singleton objects (database connections, ML models, GridSimulator instances). `@st.cache_data` caches serializable dataframes and computation results.

Chapter 4: Pandapower Physics Engine & Newton-Raphson Power Flow Math

4.1 Electrical Network Modeling

Pandapower builds 7-bus 11kV/0.4kV distribution feeder networks using `pp.create_bus()`, `pp.create_transformer_from_parameters()`, `pp.create_line_from_parameters()`, `pp.create_load()`, and `pp.create_sgen()`.

4.2 Newton-Raphson Power Flow Math

Solves non-linear nodal power balance equations $S = V * I^*$ iteratively: $J * \Delta x = \Delta f$, calculating per-unit bus voltages (V_{pu}), line loading capacities (%), and active heat losses (kW).

Chapter 5: OpenDSS Short-Circuit Impedance & Relay Trip Curves

5.1 Short-Circuit Sequence Impedance Math

Calculates line impedance $Z_{line} = dist_km * Z_{per_km}$. Evaluates positive sequence $Z1 = Z_{source} + Z_{line}$ and zero sequence $Z0$ approx $3 * Z1$.

5.2 Fault Currents & IEC 60255 Overcurrent Protection

Evaluates 3PH fault ($I_f = V_{ph} / |Z1 + Rf|$), SLG ($I_f = 3 * V_{ph} / |2 * Z1 + Z0 + 3 * Rf|$), and IEC Extremely Inverse Relay Trip Time: $t = 80 / ((I_{fault} / I_{pickup})^2 - 1)$ seconds.

Chapter 6: Scikit-Learn Machine Learning Architecture

6.1 IsolationForest Unsupervised Anomaly Detection

Partitions feature space [$V_a, V_b, V_c, I_a, I_b, I_c, PF$] using random decision trees. Outliers (voltage sags <180V) isolate near tree roots, earning high negative anomaly scores.

6.2 RandomForest Supervised Load Forecasting

Trains an ensemble of 50 decision trees mapping (hour, dayofweek) to active power demand (kW). Calculates 90% upper/lower confidence bands.

Chapter 7: SQLite Database Engine & Time-Series Persistence

7.1 Relational Storage Schema

SQLite manages smart_grid.db with 3 tables: telemetry, fault_events, and anomalies. Uses check_same_thread=False for multi-threaded thread safety.

7.2 Baseline Seeding Math

Seeds 240 historical 3-phase records spaced 6 minutes apart with daily sine wave load curves: $\text{load_factor} = 0.7 + 0.3 * \sin(\pi * (\text{hour} - 6) / 12)$.

Chapter 8: MQTT IoT Protocol & Sensor Telemetry Pipeline

8.1 MQTT Architecture

Publish-subscribe ISO standard messaging over TCP port 1883 at broker.hivemq.com. Topic path: esp32/smartgrid/telemetry.

8.2 JSON Payload Serialization

ESP32 serializes telemetry dictionaries to UTF-8 JSON text using ArduinoJson / json.dumps(), consuming minimal bandwidth.

Chapter 9: Frontend Architecture & Cyber Glassmorphism Styling

9.1 Cyber Dark CSS Layout

Injects GitHub dark background (#0d1117), 4rem container top padding, glowing blue glassmorphism metric cards, side-by-side Plotly waveforms, and fixed bottom-right 'Made by Yash' badge.

Chapter 10: Source Code Analysis & File Dependencies

10.1 File Inventory

run_system.py (launcher) -> app.py (UI) -> database.py (SQLite) -> grid_simulation.py (Pandapower) -> fault_analysis.py (OpenDSS) -> ai_analytics.py (Scikit-Learn) -> esp32_simulator.py (MQTT).

Chapter 11: System Integration & Execution Flow

11.1 Master Pipeline

run_system.py initializes SQLite database, spawns background daemon thread running esp32_simulator.py (1s telemetry loop), and launches Streamlit server.

Chapter 12: Deployment Pipeline (Streamlit Cloud & Docker)

12.1 Streamlit Cloud

Pushes repo to GitHub (yashdeep043/smart-grid-monitoring) -> Deploys at <https://smartgrid-monitoring.streamlit.app> with automatic requirements.txt installation.

Chapter 13: 60 Comprehensive Project Improvements

13.1 Roadmap

20 Beginner (CSV export, unit tests) -> 20 Intermediate (Docker containerization, PostgreSQL, JWT login) -> 20 Advanced (Hardware relay control, LoRaWAN 15km, LSTM neural network).

Chapter 14: Master Interview & Viva Q&A; Guide

14.1 100 Questions

Covers CPython internals, GIL, Pandapower Newton-Raphson math, OpenDSS impedance sequence math, IsolationForest anomaly detection, MQTT QoS, and Streamlit session state.

Chapter 15: Enterprise SCADA Standards & Industrial Scalability

15.1 Enterprise Architecture

Compares legacy Siemens SCADA (DNP3/Modbus, \$500k+) with your IoT platform (MQTT, Python, physics+AI, <\$25 per node).